Application Modernization on Power



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Modernize with speed and confidence while de-risking mission-critical with IBM Power



Protect the Core

Keep mission-critical steady while adding new capabilities.



Deployment Flexibility

Run each workload on the platform that matches its value and purpose.



Trusted, Accessible Data

Make data easy to find, current, and governed so analytics, AI, and automation deliver consistent results.



AI with Purpose

Focus AI where it improves outcomes and aligns with risk and compliance.



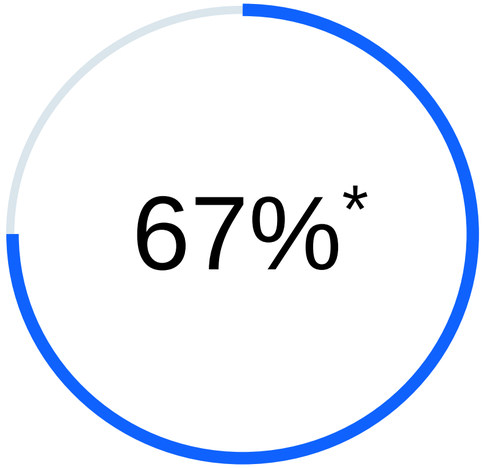
Fast, Reliable and Secure Delivery

Move fast with small releases, measurable impact, and built-in reliability and security.

Modernization must be seen as *progress*, not just *change*.

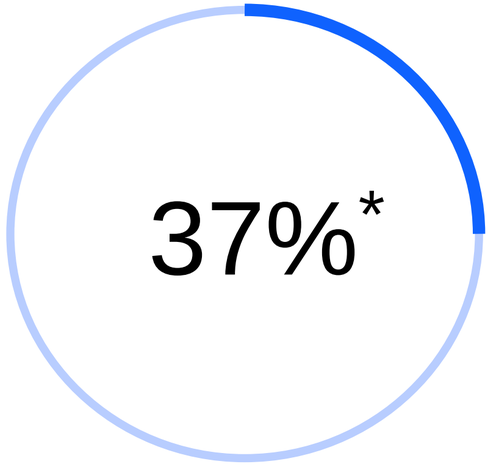


Rapid AI innovation and tech disruption is intensifying competition…



of CEOs invest early to avoid falling behind

However, caution continues to outweigh action



of CEOs are willing to move fast and learn along the way

em https://www.ibm.com/thought-leadership/institute-business-value/en-us/report/core-banking-modernization

The biggest cost isn’t modernization. It’s standing still.

The Cost of Inaction

Every year of delay adds a 25–50% maintenance tax on long standing applications.— even while the platform itself remains efficient and secure.

- **Maintenance labor:** Outdated design patterns demand 25–40% more effort per feature delivered[1]
- **Technical debt:** 10–20% of technology budgets are diverted to resolving technical debt issues each year.[2]
- **TCO impact:** Technical debt inflates TCO up to 40% of an organization's technology estate value[2]
- **Lost agility:** Enterprises running on older application architectures risk slower cycle times, missed integrations and elevated risk of falling behind.

Modernization Benefits

Application Modernization helps control costs, reduce risk and preserve business logic.

- Modernizing Applications **in place** protects core business logic while enabling automation to cut maintenance overhead.
- Incremental refactoring (modularizing, API-enabling, modernizing data) achieves **cost reduction without re-platforming**.
- Organizations using continuous modernization models see a **35% lower TCO** and **30% faster time-to-revenue** on new services (IBM Research 2024).

Every year of delay compounds technical debt, erodes business agility, and increases operational risk

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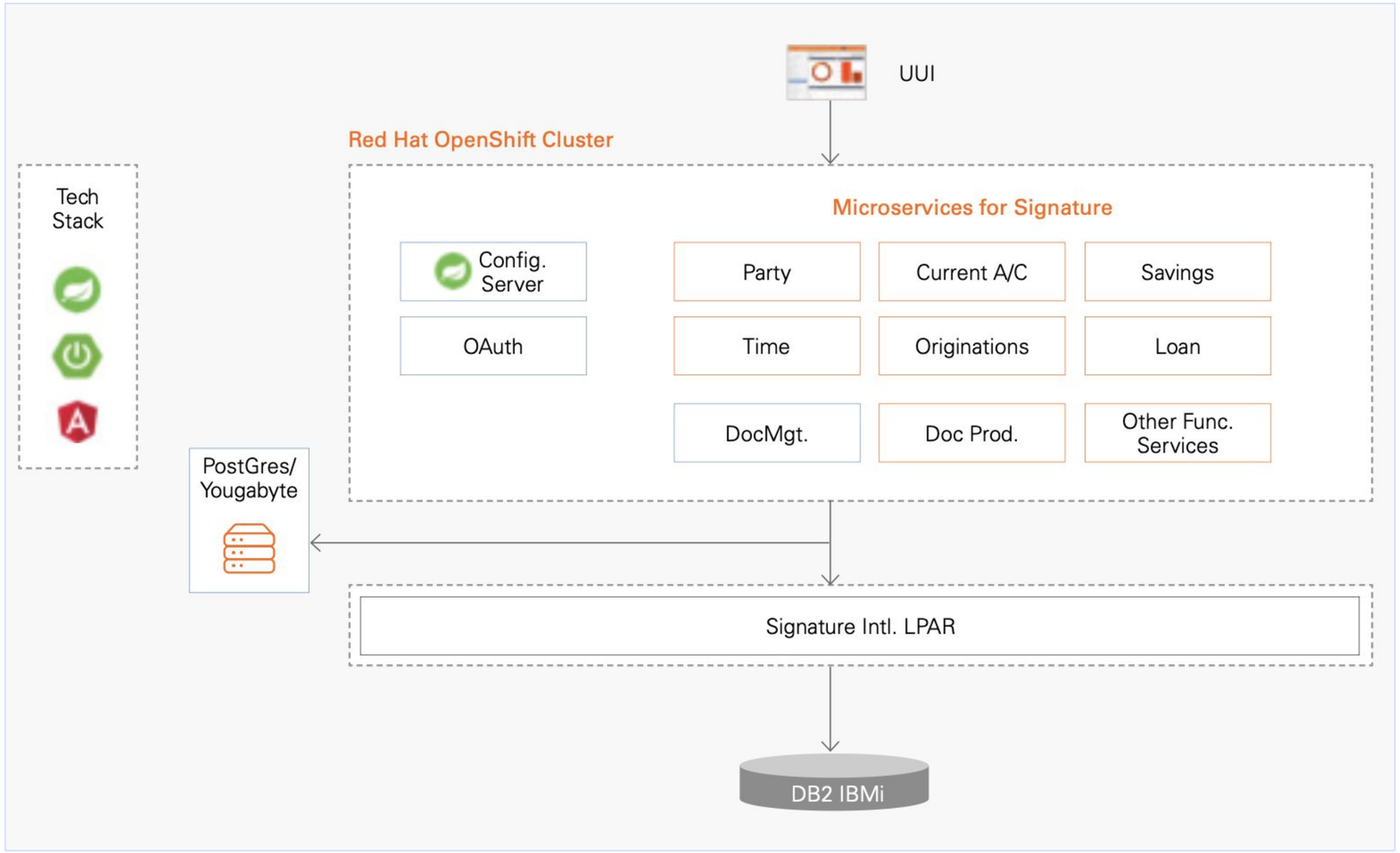


Fast, Reliable and Secure Delivery

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Modernization must be seen as *progress*, not just *change*.

A Large IBMi based Banking ISV (Fiserv) Modernizing on IBM Power



Challenge

Modernizing the Signature International Core Banking Platform without disrupting critical banking operations

Solution

Deployed Red Hat OpenShift microservices alongside Signature International workloads on IBM Power, enabling cloud-native innovation at the existing core.

Why Power?

Supports traditional and containerized workloads on one platform, accelerating modernization without core replacement.

Outcome

- à **~19K TPS achieved** in OpenShift Testing on Power
- à **Faster Delivery of new banking services**
- à **Cloud-native Modernization with minimal risk**
- à **Lower Modernization cost & time to value** while having better infra utilization and hybrid cloud flexibility

A US Bank Modernizes Payments with FTM on RH OpenShift on Power

Modernized FTM with container-native architecture – enabling real time payments, faster time-to-market and lower TCO

Key Challenges

- Legacy FTM deployments limit agility and scalability
- Need real-time payments without disrupting operations
- Manual payment repairs increase SLA risk and costs
- Compliance & security for containerized workloads
- High operational overhead and licensing costs

Solution components

- IBM Power10 S1022
- Power Enterprise Pools (PEP 2.0)
- IBM Power VC
- Red Hat OpenShift
- IBM DB2 (existing)
- IBM FTM v4 for Check (Containerized on OCP)
- Power SC & OpenShift native security

Outcomes

- ~40% faster time-to-market for payment services
- Around 35% reduction in total cost of ownership (TCO) for overall solution
- Improved resiliency and uptime (≥ 99.99%)
- Future ready hybrid cloud for AI and API-enabled architecture

Infra Readiness & Platform Optimization

FTM for check to Containers

Payment Repairs Enablement

Security & Observability

Real-time Payments & AI Ready

AI Enablement, API Augmentation and Hybrid Cloud

[Read the full story]

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Executive Summary

Customer Profile

A leading U.S. bank with nationwide presence defined by 3,600+ branches, 4,700+ ATMs, ~$650 B+ in assets; longstanding IBM Power & FTM client focused on innovation.

The Transformation

Modernized payments by upgrading IBM Power10 S1022 and adopting Red Hat OpenShift. Implemented IBM Financial Transaction Manager (FTM) for check; moved from LPARs to containers with minimal change, enabling faster releases and resilient operations while laying the groundwork for real-time payments and AI-driven fraud and hybrid cloud readiness.



How We Have Helped Organizations Navigate Modernization…



Industry Specific…

Banking & Finance

- Open Banking
- Hyper Personalization
- Fraud Management
- Digital Banking

Healthcare

- Patient Monitoring
- AI Diagnostics & Insights
- Clinical Trial Optimization
- Patient Data Integration

Retail

- Digital E-commerce
- Inventory Optimization
- Omni-channel Analytics
- POS System Modernization

Cross Industry…

DevOps | Automation…

- DevOps Enablement
- Data Modernization
- Unified IAM
- GenAI Coding Asst.

Cloud Native Modernization…

- Middleware Modernization on Cloud
- Event Driven Architectures
- Cloud Native Development
- Containerize Legacy Apps

AND Many More…..



45 +

Win wires and Public References

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IBM Power provides key capabilities that address Application Modernization challenges and accelerate digital business transformation outcomes

Ecosystem

ISVs | GSIs | BPs supporting modern applications, databases and hybrid cloud

Temenos, Finacle, IBM FTM HCL Comm.

Profound Logic, Open Legacy

GSIs

Application Modernization Foundation

Runtimes | Middleware | APIs | Modern UIs | Containers | Dev Ops | Coding Assistant

OpenShift Dev Spaces GitLab, GitHub Jenkins

IBM Cloud Paks

IBM Bob

Data Modernization Foundation

DDS to SQL | Open-source, NoSQL DBs | Data Fabric | AI Enabled (vector, etc.)

watsonx. data

EDB PostgreSQL MariaDB

MongoDB CockroachDB

Hybrid Cloud Infrastructure

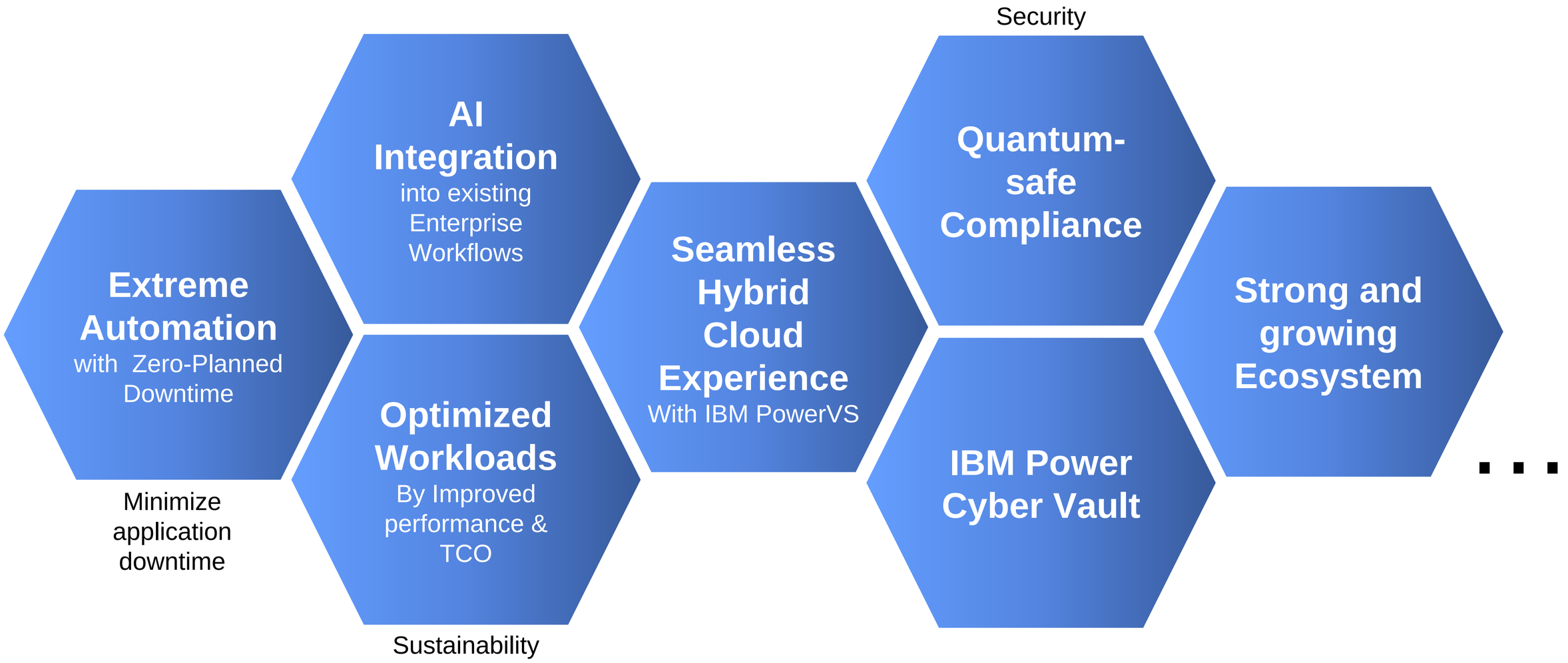
Private Cloud | Public Cloud (PVS) | Virtualization | Automation & Ops

Ansible Automation Platform

PowerVM PowerVC PEP 2.0

OCP IPI OCP MultiArch

Power 11 accelerates AI enabled Digital Business Transformation



Built on the foundation of

Performance, Availability & Security

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Power11 + Finacle Core Banking Solution

Improve Operational Efficiency & **Lower TCO**

leveraging Full Stack enhancements for

dedicated processor configurations

Infosys® | Finacle

Workload Improvements
SW Improvements *Scaling, Affinity, Efficiency*
HW Innovation *Frequency, DDR5, PHYP*

Power E1150 delivers 55% improvement

in per core database throughput performance with

25% better transaction response time

compared to Power E950

Based on IBM internal measurements for similar throughput of a commercial banking transactional database(OLTP) workload executing on 1xE1150 with 9 Power11 CPUs @ 3.5-4.2GHz compared 1xE950 with 14 Power9 CPUs @ 3.2-3.8GHz

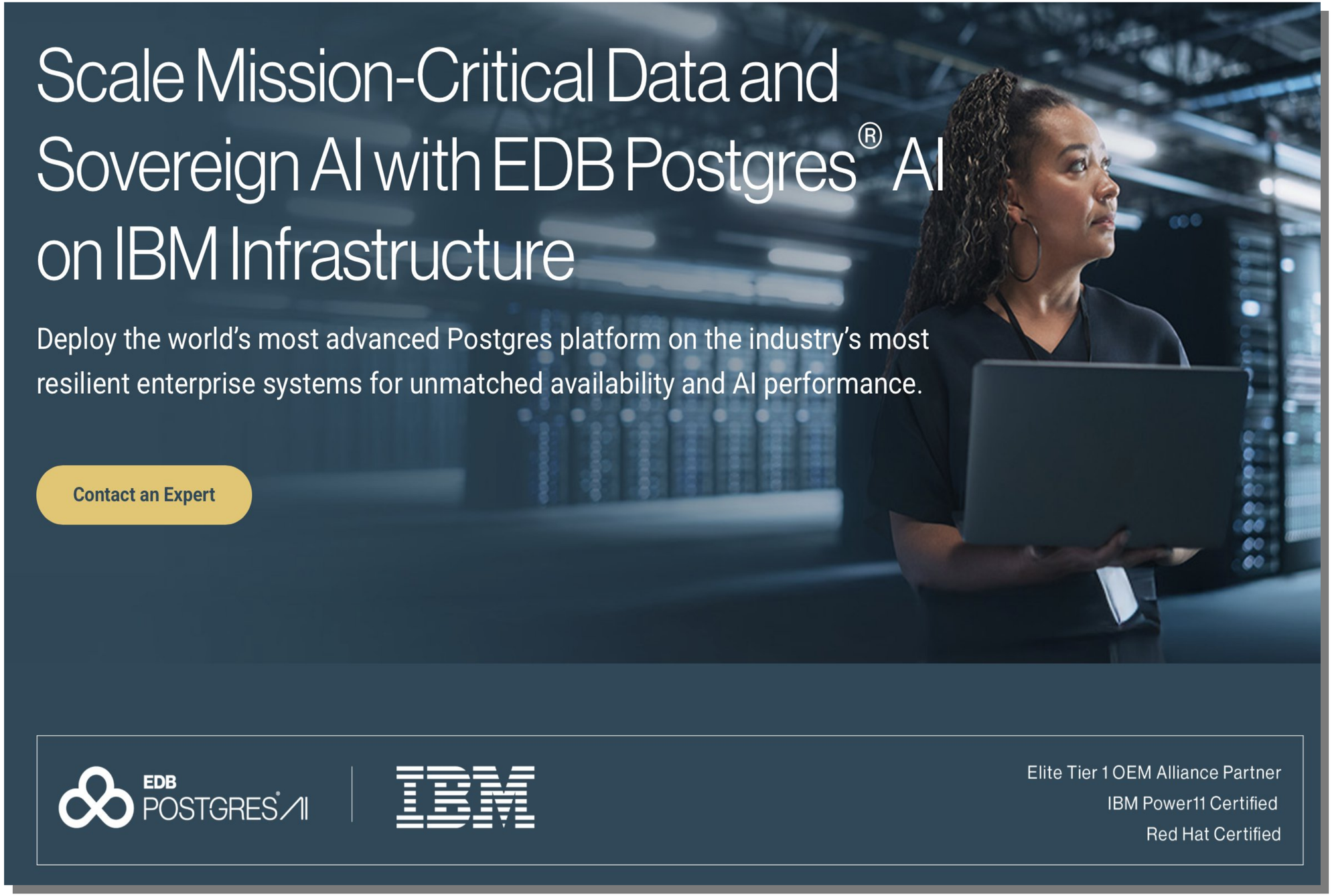
Power11 + Temenos Transact Core Banking Solution

Improved Operational Efficiency

for Temenos Transact with Red Hat OpenShift

4X performance/core running **Temenos Transact** on E1180 versus compared x86

Based on IBM internal testing for Temenos Transact core banking workload (transactions per second) accessing Linux EDB databases each running Based on IBM internal testing for Temenos Transact core banking workload (transactions per second) accessing Linux EDB databases each running 2,560,256 transactions using JMeter V5.6.3, the error ratio was kept under 1%, Results valid as of June 25, 2025 and conducted under laboratory conditions, individual results can vary based on workload size, use of storage subsystems and other conditions. Comparison is based on an IBM Power E1180 (16x16 core) with 32 worker cores versus Intel Xeon x86 (4x60 core) with 128 worker cores. Tests were run with Red Hat Enterprise Linux 9.6 on the OpenShift Container Platform Helper, Red Hat OpenShift 4.18.13 and EDB 17.5.0 on both servers.



EDB runs 2.1X faster on Power than x86

Accelerating Application Modernization on IBM Power: **Entry Points** & Innovation Paths



Modernize Infrastructure
for
Hybrid Cloud Flexibility
and improved TCO



Modernize Applications
for
Improved CX
&
Faster Time-to-Market



Modernize Data
for
Insights,
AI &
competitive advantage



Streamline Development and Operations
for
Secure, Scalable and
Faster Service Delivery

Patterns for Optimize Applications for Hybrid Cloud Flexibility and improved TCO

*Bosch witnessed **20% more energy savings** by improving the Hardware Currency*

*Sicredi gained **30% more performance** by upgrading its Hardware Currency*

Upgrade HW & SW Currency

Modernize H/W and S/W to improve performance, security and scalability without code change

Standardize virtualization, Cloud Infra Management & Cloud Like Pricing

Enable consistent operations and cost efficiency across hybrid environments with Cloud Economics

Extend to Cloud with PowerVS

Extend workloads to cloud with seamless integration and scalability

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Sicredi

One of the largest banks of ISA

BOSCH

CSI CELEBRATING 40 YEARS

celepar

Capabilities Driven Roadmap for Digital Business Transformation on Power

Currently available
On-track
Uncommitted: working it

Last Updated: 07/15/2026

	2025 & Earlier	2026	2027
Ecosystem *(link to full catalog of Industry ISVs on Linux & OpenShift Power)*	**Industry ISV Applications on OpenShift or Linux on Power**	•Temenos Transact R25, TPH •Temenos Data Hub •Finacle CMS, DEH, Finacle AI •FTM Payments - TCH, NACHA •HCL Commerce •Crest Doc. Mgmt. (Alfresco) • In10s Technologies • Diebold ATM Management • Clari5 Fraud Management • Finacle 11 (AIX) • VM 2020 • BPC SmartVista	•FTM Payments - SCT INST, Zelle, CPA •VersaFile Docuflow (ECM) •Temenos Transact R26 •Enoviq (Insurance) •Fiserv Core Advantage •Antuar Branch Innovation & Teller Soln •Aqua Global (Aquila Financial Messaging Hub) •ezbob (LendingaaS PF) •Finacle TLTM (Thin Ledger)	•Finacle Payments •FIS MBP (SDK) •ACI Worldwide (New Version)
Application Modernization Foundation *(link to complete Red Hat, IBM, ISVs & Open-source SW on Power roadmaps)* Agile Development & Ops	**Modern UI Frameworks**	•**IBM i:** Profound UI, Presto, Lansa •**AIX** Toolbox for Open-Source SW (PHP, Perl, Python, Git) •**Linux:** PHP, Angular, Javascript, node.js, Express.js	
	Runtimes & Middleware	•WebSphere, WebSphere Liberty, IBM JDK (Cloud Pak for Apps) •RH Runtimes (node.js, Springboot, Quarkus) •RH JBoss EAP, JBoss EAP XP, JBoss Web Server	•CP4Apps including RH Middleware
	APIs, Events, MQs	•IBM API Connect, App Connect, MQ, Event Streams (CP4I) •Red Hat 3 Scale, Streams for Apache Kafka, AMQ Broker •MuleSoft Flex Gateway	•Ozone API •Digibee iPaaS •IBM API Connect V12 + NanoGateway •CP4I WebMethods
	Containers & Cloud Native App Support	•OCP 4.19, 4.20 (P11 support) •OCP Platform Plus: (OCP+ACM+ACS+ODF+Quay) •OpenShift Add-ons (.NET 10, OADP, ODF) •Open-Source: Kubernetes, Docker, Prometheus, Istio, Helm	•OCP 4.21, OCP 4.22 - IPI with PowerVC •Storage Fusion 2.xx •RHCL • External Secrets Operator & Cert Mgr •RHWA (Workload Availability) • OpenShift Virtualization (KVM / Kubervirt) • Crowdstrike • OCP 5.1 / 5.2…
	DevOps & Security	•OpenShift Add-ons (GitOps, Pipelines, ODO, Service Mesh) •RH Dev Spaces, Quay, Advanced Cluster Security •Jenkins, Travis, Maven, Trilio, GitLab, Github, GH Actions, Spyderbat	•Trend Micro Container Security •Trend AI Security (CREM & XDR)
	Developer Efficiency	•Project Bob – AI Coding Asst. Public Preview	•IBM Bob Premium Package for i
Data Modernization Foundation *(link to full catalog of DBs on Linux & OCP on Power)*	**Databases – Enterprise & OSDBs**	•EDB PostgreSQL & PG Vector, Mongo DB Ent., Maria DB Ent. (OCP), MariaDB Vector, Redis, Cassandra, Milvus Vector DB, •Db2, Db2 Data Warehouse	•Mongo DB OCP Operator •Cockroach DB •eXate (Data Privacy & Security PF) •Yugabyte DB •AeroSpike DB •EDB AI Factory •DataStax (IBM D&AI OEM Version)
	Data Fabric & AI Enablement	•CP4D — Watson Pipelines, IKC, Datastage, Data Gate ext. mode • watsonx.data (Presto - Java & C++, Spark – Java) • Spyre Accelerator and SW Stack GA • Red Hat AI Inference (Spyre Only) • Red Hat OpenShift AI (OAI)	•IBM AI Launchpad services (Entity extraction, Translation) •IBM AI Launchpad UI (installation of services, ERP auto-form filling, etc.), Additional model support •OpenSearch ElasticSearch •Db2 Cloud Rack for P11 •watsonx.data (Spark gluten, Opensearch) •watsonx.data (Zero-copy data sharing with SAP BDC) •Red Hat OAI (AutoML, Auto RAG , Model serving, RAGAS, LLM guardrails, vllm-CPU) •Red Hat OAI (llm-d) •Confluent Platform •EDB PG Analytics Accelerator •EDB PD WH
Hybrid Cloud Infrastructure **--- For IBM Sellers & BPs under NDA ---**	**Private Cloud – On Prem / PVS Private**	•IBM PowerVM •IBM PowerVC, PowerVC IPI TP as part of OCP 4.21, PEP 2.0	•PowerVC IPI install auto. (GA) OCP 4.22
	Public Cloud – PVS	•PowerVS + Watsonx toolkit MVP •PowerVS Quick Start (Deployable Architecture)	•Maria DBaaS on PVS •PVS + Watsonx toolkit GA
	Automation & Ops	•Cloud Pak for Business Automation, Cloud Pak for AIOps •RH Ansible Automation Platform, RH Advanced Cluster Manager •MAS 9.1 Select Modules	•MAS Additional Modules •Concert OS Support •Terraform Provider for Power HMC •Terraform Provider PowerVC •Vault for Linux Support •Vault for AIX Support

A wide ISV ecosystem on IBM Power and many more …

Critical Applications, Open Source and OpenShift container packages for Application Modernization and Cloud Native Apps

Over 15,000 Open-source packages for IBM Power

OS Images

Red Hat, alpinelinux, CentOS, ubuntu, SUSE, debian, BusyBox, AIX, IBM i

Web & Middleware

WebSphere Liberty, JBoss by Red Hat, Open Liberty, WildFly, Apache Tomcat, ActiveMQ, RabbitMQ, Apache HTTP Server, WordPress, NGINX, Oracle Fusion Middleware

Cloud & DevOps

Red Hat OpenShift, kubernetes, Terraform, Ansible, Maven, Travis CI, Jenkins, Gradle

Languages & Runtimes

Java, python, Go, node.js, .NET, OpenJDK, Ruby, The Rust Programming Language, php, GCC, Erlang, appsody

Databases

PostgreSQL, MySQL, EDB Enterprise DB, MariaDB, mongoDB., cassandra, CouchDB, relax, redis, IBM Db2, Milvus, SAP HANA, Oracle

Analytics & AI

TensorFlow, PyTorch, ONNX, jupyter, Spark, kafka, fluentd, elasticsearch, logstash

Storage

Container Storage Interface, IBM Spectrum Virtualize, PowerVC CSI Driver, samba, NFS

Network & Monitoring

Prometheus, Nagios, etcd, Apache ZooKeeper, HAPROXY, Grafana, kibana

ISV Applications

Infosys Finacle, FIS, TEMENOS The Banking Software Company, Oracle FLEXCUBE, SWIFT, IBM FTM, IBM MAXIMO, HCL Commerce, fiserv., Epic, SAP, Oracle Applications, MUREX™

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IBM Power FSI ISV Ecosystem

Sellers can select the workloads which surround the Core Banking Application.

FSI Applications

Consumer & Community Banking (CCB)

Payments / Cards / Digital

ACI Worldwide – Universal Payments
FIS – Worldpay
Fiserv – Card & Payment Solutions
IBM – FTM
Worldline –ASSCEND
Volante – Payments Hub
Swift – FIN / ISO-20022
Temenos – Payments Hub
BPC – Smart Vista
ezbob – Digital Lending
Finacle – Lending, Payments, Engagement
Antuar - Branch Banking Software / Branch Automation Platform

Core Banking

Jack Henry – Silverlake
FIS – Profile, IBS
Fiserv – Signature
Finastra – Midas, Essence
Temenos – Transact
Oracle – FlexCube
Finacle – Core Banking
Silverlake – Silverlake Axis
Intellect Design – Core Banking

Commercial & Investment Bank (CIB)

Capital Markets / Trading

Calypso – Cross-asset front-to-back
Murex – Market Risk & Trading
FIS – GMI

Treasury / Liquidity

FIS – Treasury & Risk Management
Fiserv – Signature Treasury
Finastra – Equation
Finacle – Cash Management Suite

Wholesale Payments / Messaging

ACI Worldwide – Universal Payments
Swift – FIN / ISO-20022
Volante – Integration & Payments Hub
Temenos – Payments Hub
Finacle – Payments Suite
Aqua Global – Aquilia

Asset & Wealth Management

Wealth / Trust Accounting

FIS – GlobalPlus

Risk, Fraud & Compliance

AML / Fraud / Compliance

Actimize (NICE) – AML, Fraud, Risk
Clari5 – Fraud Management
Oracle – FS Analytical Apps
FIS – Risk & Compliance Engines
Fiserv – AML & Fraud
eXate – Data Protection / Encryption
Ozone API – Open Banking / Open Finance

Databases

Open-Source

SQL

EnterpriseDB
MariaDB
CockroachDB

NoSQL

MongoDB

Proprietary / Commercial

SQL

Db2
Oracle
SAP HANA

Lakehouse

watsonx.data
Cloudera
EDB Warehouse

Middleware

Integration / iPaaS

API Connect
MuleSoft Flex Omni Gateway
RedHat 3scale API Management
Digibee

Payments Integration

Volante
ACI / Temenos

Messaging

Swift
IBM MQ

Cloud Paks

Cloud Paks for Integration

Cloud Paks for Business Automation

Cloud Paks for Data

Cloud Paks for Watson AIOps

Security

Endpoint & XDR

Trend Micro
Fortra
PowerSC
Uptycs
Spyderbat
Mondoo

Threat & VM

Trend Micro
Qualys
Uptycs
PowerSC
Tenable Nesus

HA/DR/Obs

Veeam
Cohesity
Dynatrace
VM20/20
Trilio
IBM Concert

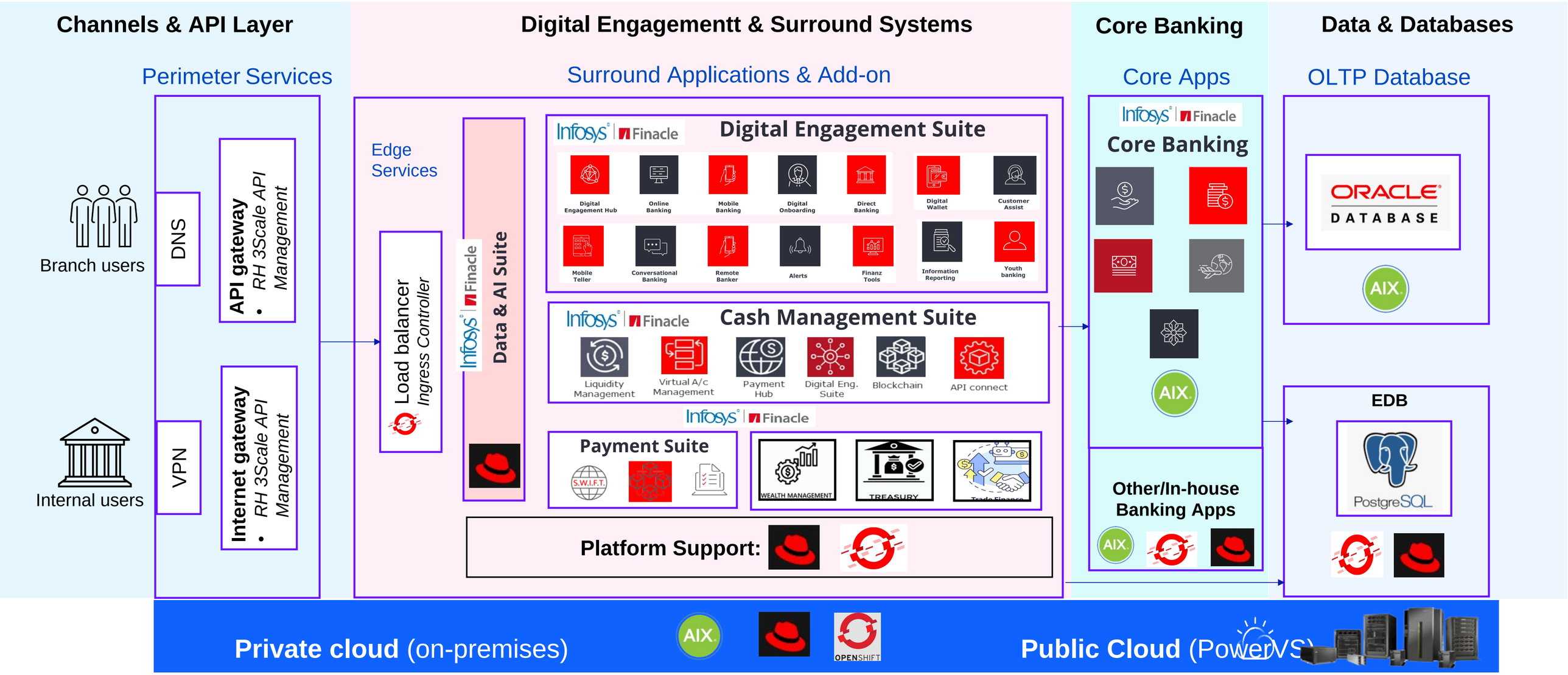
Data Security

eXate

Delivered On-Prem, Cloud, Hybrid.

Please see the appendix for operating system.

Modern Finacle Architecture on Power



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